



GABAPIN

(Gabapentin Tablets 600 mg)

Name and strength of active ingredient
Gabapentin 600

Dosage form
Film Coated Tablet

Therapeutic Code
Antiepileptics: ATC code NO3AX12

Product Description

Gabapentin Tablets are available in two strengths: 600 mg
GABAPIN 600 Tablets is White to off white, oval shaped, film coated tablets plain on both sides.

Pharmacodynamic properties

Gabapentin is structurally related to the neurotransmitter GABA (gammaaminobutyric acid) but its mechanism of action is different from that of several other drugs that interact with GABA synapses including valproate, barbiturates, benzodiazepines, GABA transaminase inhibitors, GABA uptake inhibitors, GABA agonists, and GABA prodrugs.

Pharmacokinetic properties

Absorption

Following oral administration, peak plasma gabapentin concentrations are observed within 2 to 3 hours. Gabapentin bioavailability (fraction of dose absorbed) tends to decrease with increasing dose. Absolute bioavailability of a 300 mg capsule is approximately 60%. Food, including a high-fat diet, has no clinically significant effect on gabapentin pharmacokinetics.

Gabapentin pharmacokinetics is not affected by repeated administration. Although plasma gabapentin concentrations were generally between 2 µg/ml and 20 µg/ml in clinical studies, such concentrations were not predictive of safety or efficacy. Pharmacokinetic parameters are given in below Table.

Table - Summary of gabapentin mean (%CV) steady-state pharmacokinetic parameters following every eight hours administration

Pharmacokinetic parameter	300 mg (N = 7)		400 mg (N = 14)		800 mg (N=14)	
	Mean	%CV	Mean	%CV	Mean	%CV
C _{max} (µg/ml)	4.02	(24)	5.74	(38)	8.71	(29)
t _{max} (hr)	2.7	(18)	2.1	(54)	1.6	(76)
T _{1/2} (hr)	5.2	(12)	10.8	(89)	10.6	(41)
AUC (0-8) µg.hr/ml	24.8	(24)	34.5	(34)	51.4	(27)
Ae% (%)	NA	NA	47.2	(25)	34.4	(37)

C_{max} = Maximum steady state plasma concentration

t_{max} = Time for C_{max}

T_{1/2} = Elimination half-life

AUC(0-8) = Steady state area under plasma concentration-time curve from time 0 to 8 hours postdose

Ae% = Percent of dose excreted unchanged into the urine from time 0 to 8 hours postdose

NA = Not available

Distribution

Gabapentin is not bound to plasma proteins and has a volume of distribution equal to 57.7 litres. In patients with epilepsy, gabapentin concentrations in cerebrospinal fluid (CSF) are approximately 20% of corresponding steady-state trough plasma concentrations. Gabapentin is present in the breast milk of breast-feeding women.

Metabolism

There is no evidence of gabapentin metabolism in humans. Gabapentin does not induce hepatic mixed function oxidase enzymes responsible for drug metabolism.

Elimination

Gabapentin is eliminated unchanged solely by renal excretion. The elimination half-life of gabapentin is independent of dose and averages 5 to 7 hours.

In elderly patients, and in patients with impaired renal function, gabapentin plasma clearance is reduced. Gabapentin elimination-rate constant, plasma clearance, and renal clearance are directly proportional to creatinine clearance.

Gabapentin is removed from plasma by haemodialysis. Dosage adjustment in patients with compromised renal function or undergoing haemodialysis is recommended.

Gabapentin pharmacokinetics in children was determined in 50 healthy subjects between the ages of 1 month and 12 years. In general, plasma gabapentin concentrations in children > 5 years of age are similar to those in adults when dosed on a mg/kg basis.

Linearity/Non-linearity

Gabapentin bioavailability (fraction of dose absorbed) decreases with increasing dose which imparts non-linearity to pharmacokinetic parameters which include the bioavailability parameter (F) e.g. Ae%, CL/F, Vd/F. Elimination pharmacokinetics (pharmacokinetic parameters which do not include F such as CL_r and T_{1/2}), are best described by linear pharmacokinetics. Steady state plasma gabapentin concentrations are predictable from single-dose data.

Indication/Usage

Epilepsy

Gabapentin is indicated as adjunctive therapy in the treatment of partial seizures with and without secondary generalization in adults and children age 3 years and above.

Safety and effectiveness for adjunctive therapy in pediatric patients below the age of 3 years have not been established.

Neuropathic pain

Gabapentin is indicated for the treatment of neuropathic pain which includes diabetic neuropathy, post-herpetic neuralgia and trigeminal neuralgia in adult age 18 years and above.

Safety and effectiveness in patients below the age of 18 years have not been established.

Recommended Dose

General

Gabapentin is given orally with or without food.

When in the judgment of the clinician there is a need for dose reduction, discontinuation, or substitution with an alternative medication, this should be done gradually over a minimum of one week.

Epilepsy

Adults and pediatric patients over 12 years of age

In clinical trials, the effective dosing range was 900 to 3600 mg/day. Therapy may be initiated by administering 300 mg three times a day (TID) on Day 1 or by titrating the dose as described in Table 1. Thereafter, the dose can be increased in three equally divided doses up to a maximum dose of 3600 mg/day. Dosages up to 4800 mg/day have been well tolerated in long-term open-label clinical studies.

The maximum time between doses in the three times a day (TID) schedule should not exceed 12 hours to prevent breakthrough convulsions.

Table 1 DOSING CHART - INITIAL TITRATION		
Day 1	Day 2	Day 3
300 mg once a day	300 mg two times a day	300 mg three times a day

Pediatric patients age 3-12 years

The starting dose should range from 10 to 15 mg/kg/day given in equally divided doses (three times a day), and the effective dose reached by upward titration over a period of approximately three days. The effective dose of gabapentin in patient age 5 years and older is 25 to 35 mg/kg/day given in equally divided doses (three times a day). The effective dose in pediatric patients ages 3 to less than 5 years is 40 mg/kg/day given in equally divided doses (three times a day). Dosages up to 50 mg/kg/day have been well tolerated in a long-term clinical study. The maximum time interval between doses should not exceed 12 hours.

It is not necessary to monitor gabapentin plasma concentrations to optimize gabapentin therapy. Further, gabapentin may be used in combination with other antiepileptic drugs without concern for alteration of the plasma concentrations of gabapentin or serum concentrations of other antiepileptic drugs.

Neuropathic pain in Adults

The starting dose is 900 mg/day given as three equally divided doses and increased if necessary, based on response, up to a maximum dose of 3600 mg/day. Therapy should be initiated by titrating the dose as described in Table 1.

Dosage adjustment in impaired renal function for patients with neuropathic pain or epilepsy

Dosage adjustment is recommended in patients with compromised renal function as described in Table 2 and/or those undergoing haemodialysis.

Table 2 DOSAGE OF GABAPENTIN IN ADULTS BASED ON RENAL FUNCTION	
Creatinine Clearance (ml/min)	Total Daily Dose ^a (mg/day)
80	900-3600
50-79	600-1800
30-49	300-900
15-29	150 ^b -600
<15	150 ^b -300

^a Total daily dose should be administered as a TID regimen. Doses used to treat patients with normal renal function (creatinine clearance >80 ml/min) range from 900 to 3600 mg/day. Reduced dosages are for patients with renal impairment (creatinine clearance < 79 ml/min).

^b To be administered as 300 mg every other day.

Dosage adjustment in patients undergoing haemodialysis

For patients undergoing haemodialysis who have never received gabapentin, a loading dose of 300 to 400 mg is recommended, then 300 mg of gabapentin following each 4 hours of haemodialysis.

Route of Administration

Oral

Contraindication

Hypersensitivity to the active substance or to any of the excipients.

Warnings & Precaution

Potential for an increase in risk of suicidal thoughts or behaviours

General

Although there is no evidence of rebound seizures with gabapentin, abrupt withdrawal of anticonvulsant in epileptic patients may precipitate status epilepticus. Gabapentin is not generally considered effective in the treatment of absence seizures. Patients who require concomitant treatment with morphine may experience increases in gabapentin concentrations. Patients should be carefully observed for signs of CNS depression, such as somnolence, and the dose of gabapentin or morphine should be reduced appropriately.

Respiratory depression

Gabapentin has been associated with severe respiratory depression. Patients with compromised respiratory function, respiratory or neurological disease, renal impairment, concomitant use of central nervous system (CNS) depressants and the elderly might be at higher risk of experiencing this severe adverse reaction. Dose adjustments might be necessary in these patients.

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Information for patients

To assure safe and effective use of gabapentin, the following information and instructions should be given to patients:

1. Inform your physician about any prescription or nonprescription medications, alcohol, or drugs you are now taking or plan to take during your treatment with gabapentin.
2. You should inform your physician if you are pregnant, or if you are planning to become pregnant, or if you become pregnant while you are taking gabapentin.
3. Gabapentin is excreted in human milk, and the effect on the nursing infant is unknown. You should inform your physician if you are breast feeding an infant.
4. Gabapentin may impair your ability to drive a car or operate potentially dangerous machinery. Until it is known that this medication does not affect your ability to engage in these activities, do not drive a car or operate potentially dangerous machinery.
5. You should not allow more than 12 hours between gabapentin doses to prevent breakthrough convulsions.

Interaction with other medicaments

In a study involving healthy volunteers (N=12), when a 60 mg controlled-release morphine capsule was administered 2 hours prior to a 600 mg gabapentin tablet, mean gabapentin AUC increased by 44% compared to gabapentin administered without morphine. Therefore, patients should be carefully observed for signs of CNS depression, such as somnolence, and the dose of gabapentin or morphine should be reduced appropriately.

No interaction between gabapentin and phenobarbital, phenytoin, valproic acid, or carbamazepine has been observed.

Gabapentin steady-state pharmacokinetics is similar for healthy subjects and patients with epilepsy receiving these anti-epileptic agents.

Coadministration of gabapentin with oral contraceptives containing norethindrone and/or ethinyl estradiol, does not influence the steady-state pharmacokinetics of either component.

Coadministration of gabapentin with antacids containing aluminium and magnesium, reduces gabapentin bioavailability up to 24%. It is recommended that gabapentin be taken at the earliest two hours following antacid administration.

Renal excretion of gabapentin is unaltered by probenecid.

A slight decrease in renal excretion of gabapentin that is observed when it is coadministered with cimetidine is not expected to be of clinical importance.

Pregnancy and Lactation

Risk related to epilepsy and antiepileptic medicinal products in general

The risk of birth defects is increased by a factor of 2 – 3 in the offspring of mothers treated with an antiepileptic medicinal product. Most frequently reported are cleft lip, cardiovascular malformations and neural tube defects. Multiple antiepileptic drug therapy may be associated with a higher risk of congenital malformations than monotherapy; therefore it is important that monotherapy is practiced whenever possible. Specialist advice should be given to women who are likely to become pregnant or who are of childbearing potential and the need for antiepileptic treatment should be reviewed when a woman is planning to become pregnant. No sudden discontinuation of antiepileptic therapy should be undertaken as this may lead to breakthrough seizures, which could have serious consequences for both mother and child. Developmental delay in children of mothers with epilepsy has been observed rarely. It is not possible to differentiate if the developmental delay is caused by genetic, social factors, maternal epilepsy or the antiepileptic therapy.

Risk related to gabapentin

There are no adequate data from the use of gabapentin in pregnant women.

Studies in animals have shown reproductive toxicity. The potential risk for humans is unknown.

Gabapentin should not be used during pregnancy unless the potential benefit to the mother clearly outweighs the potential risk to the foetus.

No definite conclusion can be made as to whether gabapentin is associated with an increased risk of congenital malformations when taken during pregnancy, because of epilepsy itself and the presence of concomitant antiepileptic medicinal products during each reported pregnancy.

Gabapentin is excreted in human milk. Because the effect on the breast-fed infant is unknown, caution should be exercised when gabapentin is administered to a breast-feeding mother.

Gabapentin should be used in breast-feeding mothers only if the benefits clearly outweigh the risks.

Side effects/Adverse Reactions

The adverse reactions observed during clinical studies conducted in epilepsy (adjunctive and monotherapy) and neuropathic pain have been provided in a single list below by class and frequency (very common (> 1/10), common (>1/100, <1/10), uncommon (>1/1000, <1/100) and rare (>1/10,000; <1/1,000). Where an adverse reaction was seen at different frequencies in clinical studies, it was assigned to the highest frequency reported.

Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

Infections and infestations

Very Common: Viral infection

Common: Pneumonia, respiratory infection, urinary tract infection, infection, otitis media

Blood and the lymphatic system disorders

Common: leucopenia

Rare: thrombocytopenia

Immune system disorders

Rare: allergic reactions (e.g. urticaria)

Metabolism and Nutrition Disorders

Common: anorexia, increased appetite

Psychiatric disorders

Common: hostility, confusion and emotional lability, depression, anxiety, nervousness, thinking abnormal

Rare: hallucinations

Nervous system disorders

Very Common: somnolence, dizziness, ataxia,

Common: convulsions, hyperkinesias, dysarthria, amnesia, tremor, insomnia, headache, sensations such as paresthesia, hypaesthesia, coordination abnormal, nystagmus, increased, decreased, or absent reflexes

Rare: movement disorders (e.g. choreoathetosis, dyskinesia, dystonia)

Eye disorders

Common: visual disturbances such as amblyopia, diplopia

Ear and Labyrinth disorders

Common: vertigo

Rare: tinnitus

Cardiac disorders

Rare: palpitations

Vascular disorder

Common: hypertension, vasodilatation

Respiratory, thoracic and mediastinal disorders

Common: dyspnoea, bronchitis, pharyngitis, cough, rhinitis

Rare: respiratory depression

Gastrointestinal disorders

Common: vomiting, nausea, dental abnormalities, gingivitis, diarrhea, abdominal pain, dyspepsia, constipation, dry mouth or throat, flatulence

Rare: pancreatitis

Hepatobiliary disorders

Rare: hepatitis, jaundice

Skin and subcutaneous tissue disorders

Common: facial oedema, purpura most often described as bruises resulting from physical trauma, rash, pruritus, acne

Rare: Stevens-Johnson syndrome, angioedema, erythema multiforme, alopecia

Musculoskeletal, connective tissue and bone disorders

Common: arthralgia, myalgia, back pain, twitching

Renal and urinary disorders

Common: incontinence

Rare: acute renal failure

Reproductive system and breast disorders

Common: impotence

General disorders and administration site conditions

Very Common: fatigue, fever

Common: peripheral or generalized oedema, abnormal gait, asthenia, pain, malaise, flu syndrome

Rare: withdrawal reactions (mostly anxiety, insomnia, nausea, pains, sweating), chest pain. Sudden unexplained deaths have been reported where a causal relationship to treatment with gabapentin has not been established.

Investigations

Common: WBC (white blood cell count) decreased, weight gain

Rare: Blood glucose fluctuations in patients with diabetes, elevated liver function tests

Injury and poisoning

Common: accidental injury, fracture, abrasion

Under treatment with gabapentin cases of acute pancreatitis were reported. Causality with gabapentin is unclear.

Respiratory tract infections, otitis media, convulsions and bronchitis were reported only in clinical studies in children. Additionally, in clinical studies in children, aggressive behaviour and hyperkinesias were reported commonly.

Post-marketing experience: Dysphagia

Signs & Symptoms of overdose and Treatment

Acute, life-threatening toxicity has not been observed with gabapentin overdoses of up to 49 grams.

Symptoms of the overdoses included dizziness, double vision, slurred speech, drowsiness, lethargy and mild diarrhoea. All patients recovered fully with supportive care. Reduced absorption of gabapentin at higher doses may limit drug absorption at the time of overdosing and, hence, minimise toxicity from overdoses.

Although gabapentin can be removed by haemodialysis, based on prior experience it is not usually required. However, in patients with severe renal impairment, haemodialysis may be indicated.

An oral lethal dose of gabapentin was not identified in mice and rats given doses as high as 8000 mg/kg. Signs of acute toxicity in animals included ataxia, labored breathing, ptosis, hypoactivity, or excitation.

Storage Conditions

Gabapentin Tablets should be kept in a safe place where children cannot reach them. Do not store above 30°C. Do not use the tablets after the expiry date printed on the packaging.

Shelf life

24 months

Pack size

Gabapentin Tablets 600 mg are packed in clear transparent PVC/PPVdC-Aluminium blister of 10 x 10 Tablets.

Manufactured by:



INTAS PHARMACEUTICALS LTD.

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Marketing Authorization Holder:

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